

Aaron Dutta

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EDUCATION

University of Waterloo <i>BMath, Honours Mathematics with Co-Op</i>	Sep 2026 - Apr 2031 <i>Waterloo, ON</i>
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SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL
Libraries/Frameworks: React.js, Next.js, Node.js, FastAPI, RestAPIs, Flask, Prisma, Tailwind CSS
AI/ML & Data: OpenCV, MediaPipe, NumPy, Pandas, Anthropic API, ONNX Runtime, NetworkX, Pydantic
Tools & Platforms: Git, Docker, PostgreSQL, Microsoft Azure, Vercel, Supabase, Figma

EXPERIENCE

Mathematics Mentor <i>Felix School of Education</i>	Nov 2024 – Aug 2026 <i>Purulia, India (Remote)</i>
- Mentoring 15+ Grade 7–12 students through weekly math circles, encouraging collaborative problem solving - Advancing students’ critical thinking skills by combining theoretical concepts with practical exercises	

PROJECTS

TechniqueTitan <i>Python, MediaPipe, OpenCV, NumPy, FastAPI, Pandas, Matplotlib</i>  	Jul 2026 - Present
- Built an AI piano posture coach that turns any device camera into on-demand technique feedback so students can correct hand form without a teacher - Engineered a MediaPipe + OpenCV + NumPy pipeline to extract 21 landmarks per hand, normalize for camera distance, and score 5 posture criteria - Shipped a FastAPI + React/TypeScript app with on-device browser MediaPipe and YAML-tunable thresholds, cutting review from a full lesson cycle to seconds - Built a batch CLI that walks image folders and exports CSV/JSON metrics along with a 45-test pytest suite across geometry, scoring, and API modules, automating dataset-scale analysis that was previously manual	
SkyEye <i>Python, Flask, YOLOv8, ONNX Runtime, React, TypeScript, Gemini API</i>  	Aug 2026
- Won RescueHacks 2026 for an SAR triage tool that turns a free-text missing-person report into a Lost Person Behavior search ring, scoring map pins in 5–8 s within a 30 s time-to-first-lead target - Developed a sliding-window YOLOv8n + ONNX Runtime pipeline to tile images into overlapping windows - Shipped a Flask + React/TypeScript operator console with Gemini Flash (Groq fallback) for typed extraction and Google Geocoding + LPB $\sqrt{t}$ radius	
FraudGen <i>Python, Anthropic API, FastAPI, Streamlit, Pandas, NetworkX, Pydantic</i>  	Mar 2026
- Engineered a 6-agent synthetic fraud data generator that closes the labeled-data gap in ML fraud detection by simulating adversarial fraud networks - Integrated Anthropic API to label fraud variants in parallel, cutting generation from 40 s to under 6 s. - Shipped an 8-rule validation system with an LLM critic, FastAPI backend, and Next.js/React dashboard, achieving under a 10% variant revision rate.	

AWARDS

• Winner @ RescueHacks	Sep 2026
• 2nd Place Overall @ Katy Youth Hacks	Aug 2026
• 3rd Place Security/Privacy Hack @ STEMist Hacks	Aug 2026
• 1st Place @ STEMMINDS Hackathon	Mar 2026

CERTIFICATIONS

• Essentials with Azure Fundamentals, Microsoft	Aug 2026
• Building with the Claude API, Anthropic	Jul 2026
• Google AI Essentials Specialization, Google	Jul 2026
• CS50x, Harvard University	Dec 2024